

ActInf Livestream #038.1 ~ "The evolution of brain architectures for predic

Livestream | 1:54:49

All right.

Hello and welcome everyone to ActInfLab livestream number 38.1.

It's February 16th, 2022.

Welcome to the ActInfLab.

We are a participatory online lab that is communicating, learning, and practicing applied active inference.

You can find us at the links on this slide.

This is a recorded and an archived livestream, so please provide us with feedback so that we can improve our work.

All backgrounds and perspectives are welcome, and we'll be following video etiquette for livestream.

Just release an unending torrent of emojis if you have to speak, or just raise your hand.

I'm sure that we'll get to it.

If you're watching live, please feel free to write questions in the live chat,

and we'll have enough time to hang out and discuss during this .1 where we'll be opening into the paper.

Check out activeinference.org for updated information on participating in any of the lab's activities.

I hope you'll find something that resonates with you.

Today in ActInf livestream number 38.1, we're going to be learning and discussing this cool paper,

The Evolution of Brain Architectures for Predictive Coding and Active Inference,

by Pizzullo, Parr, and Friston from December 2021.

And we're just going to enjoy discussing it and opening up any ideas or questions that us here on the panel have,

or those who are live chatting us.

And we have some ideas and thoughts prepared, a few things that we know that we can go into.

And also, I hope that everyone has brought some other prepared seeds,

and also, of course, spontaneously feeling like new things are arising.

So we'll just start with some introduction and warm-up.

We can each say hi, and then maybe it'd be cool to just also mention, like,

what got you excited about the paper, or what made you want to discuss this paper?

What's something that stayed with you?

So, I'm Daniel.

I'm a researcher in California, and I was very excited by the evolutionary focus.

A lot of my research over the last years has been in evolution and ecology.

So it's always awesome to see how people are thinking about how Active Inference,

Free Energy Principle, and evolutionary studies can all be learning from one another.

And I'll pass to Stephen.

Hello, I'm Stephen Sillett.

I'm in Toronto.

I'm really interested in how this paper connects to my work with spatial meaning-making and social topographies, because this paper talks about a more biological, nascent stage of development, rather than some higher order meaning, often people think about.

And I'm really interested in how this sort of grounded, bottom-up, active ecological approaches can be thought about as something where we can actually ground a lot of Armenian-making.

So I'm interested in how this might match with some of that thinking.

And I will pass this over to Dean.

Thanks, Stephen.

I'm Dean.

I'm in Calgary.

What I found interesting about the paper is that, given that I've worked with a lot of young learners for a lot of young learners, for a lot of young learners, for a lot of my life,

I was really, it was really interesting to see how this affirmed a lot of the thinking that I was doing, when I was trying to get people past the idea that science is only about the biology and the chemistry and the physics,

physics, that there is a statistical and predictive component to this.

And those relationships in that realm can build certain sense of what the underlying architecture is.

So, as I said, a lot of this is affirmations.

And it's kind of seen as being not part of what is typically addressed as core learning, but I think it should be.

Back to you, Dan.

Dan.

Cool.

Nice intro.

Let's just start with a big question.

And then either of you, if you have any reflections on the big question, or we'll go over to blank slide and just bring up some questions we have.

And also, it might be good to go over just some of the key points in the paper, like what did they actually do in terms of their contribution?

So, the big question, or at least one way to phrase something that might bring someone to approach this paper.

What is the evolutionary neurophysiological basis of cognition?

And how do complex cognitive phenotypes arise?

Like you don't go from zero to colony in one time step.

How does it happen that evolution arises from precursor, sometimes simpler, but also sometimes

more complicated precursors?

Stephen mentioned like what is the basis for sense making and cognition and sense making are very related.

And what's been fun over these last weeks and months is we've explored basic or simple or reduced or basal cognition from a variety of perspectives.

Like we talked about the bioelectric components of thinking about basal cognition with Mike Levin.

Now we're thinking about a slightly different approach to understanding basal cognition, which we'll be discussing here.

And then also we've looked at some of these more complex cognitive phenotypes like mental action, counterfactuals, deep temporal inference, all the abductive logic, which we'll get into probably later.

How can we use integrative models of perception, cognition, action, and impact like active inference to study this whole continuum and diversity of relatively simpler in our perspective cognition and also relatively more complex cognitive phenotypes and everything in between?

So yes, Stephen, and then on with it.

Thanks, Daniel.

Yeah.

I mean, I suppose also what some of the paradigm impacts of that, of taking that all the way through.

And one question that might come up, which you may or may not have an answer to or thoughts on, but Mike Levin's work with the teleological colons at the different nested levels and how this could be thinking about what kind of action control potential in these different nested levels might exist.

I would be interested to think about how this may either be able to connect to that.

I can see it connecting philosophically or whether there's another bridge needed between this modeling and that type of modeling.

So it's just something I've mentioned.

Great.

Dean, anything to add or perhaps?

I'm going to wait till you get to figure one.

Okay.

So we won't go through all of the finer points in the paper.

That's for the reading of the paper itself.

Nor will we even go over all of the overview, which is what the dot zero video 38.0 is for.

So, you know, turn back time and watch that one if you haven't or pause the video if you're not watching it live.

But what this paper does, as evidenced by their roadmap, is introduce a few basic principles of cognition.

Predictive regulation and control, and then also structure learning in generative models.

That's not how every neurophysiology text is going to begin the building blocks of cognition.

It's not how every evolutionary neurophysiology text is going to be framing cognition, but that's what they do here.

And that will play into how it's similar and different than other approaches.

Then with those building blocks in hand or on the floor, they provide three examples of motifs that ancestral brains may be modeled as having or may have actually had.

With those three examples in hand, it's then possible for them to state a more general way of thinking about the transitions amongst different structures.

These structures represent brain design as structure learning in generative models.

That is called an evolutionary algebra.

And they introduce five operators that can basically either leave unchanged or change the structure of brain design with respect to generative models.

And each of those are explored in terms of what are the architectural changes that that evolutionary transition is.

And then what are the functional consequences of that kind of an evolutionary transition?

Then they have some discussions about how sequences or patterns of application of this evolutionary algebra could lead to different evolutionary phenomena.

So, for example, increasing temporal depth in the future means that models are increasingly prospective.

Increasing temporal depth in the past is like memory, etc.

And then they close with mapping to a phylogenetic tree and thinking about that evolutionary algebra of state transitions being mapped onto the bifurcating tree structure that's called a phylogenetic tree, which shows the relatedness of different life forms.

That's the roadmap.

Let's go to the long-awaited figure one and talk about the action perception cycle and predictive regulation.

And so they're again discussing this in the context of predictive regulation, anticipatory regulation, cybernetics, and control as a basic design principle for the brain.

So, Dean, what do you see here or what would be cool to think about?

So what this follows, I think, is the usual pattern when we're trying to explain where we're going to go.

And so you just had a roadmap up.

And I think a roadmap is a way for the people who are reading the roadmap to find their way.

What I think what this diagram shows is that in the middle of there, there's something called a discrepancy.

And that discrepancy is later going to be given a label Y.

And what I think is really interesting here is there's a flip potentially that can happen

here.

That discrepancy is where what I would describe as a rule factory.

And rule in the sense that we find pattern.

That's kind of our place where, as it says, the prediction and the observation come together.

And I believe that that's different than a roadmap, which is find your way.

I believe that that discrepancy or that rule factory is wayfinding.

And I think what we have the potential to do with this paper, because of the way that they have sort of presented the information, is the words eventually become explicated rules, which then become action in the phenomenological space.

That's where it gets really, really interesting, because I think most of the time when people look at things as a subject, they've taken the world and they've collapsed down to words and diagrams and models.

What this potentially allows us to do is flip that and move from the words back out into the space with a little bit more confidence.

So, yeah, that's why I wanted to kind of start here, because I think the word discrepancy, it's the first time I've seen it in this kind of figure model, and I really like it.

Cool.

Yeah, it makes me think about setting off on the road trip, on the mental actions that reflect the paper.

And there's the roadmap, which is super informative, but it is like instructionalism.

It's saying you're going to go two streets and then take a left turn at the stop sign.

And then when this happens, then you'll do that.

And if you've seen this, then you've gone too far.

Kind of classic instruction type sequences.

Here we have a figuring out, because it's visually arranged with some local connectivity that's suggestive of a causal connection through time.

But by no means is there only one way to read this simultaneous figure.

And so that allows potentially for more of a figuring out, including a figuring out of rules.

Yes, it is indeed a little different with discrepancy at the intersection here.

Stephen?

Yeah, so this ties in with the lower bound evidence control approach of action.

So, you know, the idea is that action is what we can tractably approximate.

And perception is something that we can access but can't, and we can try to make more sense of, but is a harder piece of the equation to get a handle on.

So, I'm wondering if that's, I mean, how do you feel about that separation in terms of its use across other areas of applied active inference?

So, I think it could make things a little bit more digestible for a number of contexts.

Okay, I'm thinking about this in the context of recognizing that it's different than other action perception loops

that we've seen,

which is just really important to keep in mind that we're not perceiving something that we're projecting too much.

Because other times, the outgoing arrow from the entity is what?

If this were a Markov blanket type diagram, which they often are,

the outgoing arrow would reflect active states and then the statistical dependencies that are outgoing.

But where is action?

It's on the bottom right.

So, let's really try to understand why the pieces are placed this way.

The outgoing feature of the cognitive entity is the prediction.

And here's observation coming in.

If this were a Markov blanket diagram,

we'd have the world and then the incoming statistical arrow would be the observation.

That would be the sensory states.

So, here, it's the outgoing prediction of the entity and the incoming sensory data or observation.

Those two are being differentiated to form a discrepancy,

which is just a qualitative term,

but it could be then brought a little bit more formally into a prediction error

or a little bit even beyond that into like a free energy differential.

This discrepancy has two arrows coming out of it.

From the discrepancy is arising perception and the changing of beliefs.

Does perception always involve changing beliefs?

And discrepancy is also giving rise to action.

So, what kind of a thing is discrepancy such that the inputs are prediction and observation

and the outputs are perception and action?

So, at the very least, this is not the Bayesian graph representations that we've seen before

or that we'll see in just a few slides

with a more traditional interpretation of nodes as random variables

and edges as statistical dependencies.

This is a little bit more like a thought map

that then connects to the variational free energy equation,

which we talked about in number 37 a lot more.

But just to recall, there's the red and the blue lines,

these two different components of the variational free energy,

and those are shown again here.

So, check out 37 to learn more about the red and the blue

and about variational and expected free energy.

But for here, we're starting with this action, prediction, discrepancy motif

and then connecting it to perception and action
as variational free energy minimization.

Stephen?

I think one thing that's useful with this more nascent representation
is discrepancy can go in different directions.

So, it can be discrepancy in terms of temporal occurrence.

When was the prediction predicted to happen?

When was it observed?

But it could also be what kind of prediction
and what kind of observation or whereabouts was the prediction?

Whereabouts was the observation?

There's many different and at different scales, you know.

So, there may be that at the kind of lower levels,
there's quite a big jump between when I predict the baseball coming to my hand,
when that prediction was made, when that observation occurs,
and also when that is chained up at different, slower, bigger steps
of the nested Markov blanket sequence.

So, this idea of discrepancy is, I suppose,
it's probably the biggest bucket they could find, I would imagine,
at that spot.

And that may be partly why it's there.

Can I just add to that, Stephen?

I think one of the things that's interesting,
especially after just doing the 37 paper,
where we were talking about guides,
to me, the timing of this was absolutely immaculate
because now all of a sudden, we've gone from guiding
to almost taking, potentially, a referee's position on the world,
which is a whole different thing than taking up the position of being a guide.
And that was, I mean, that's not nuanced.

That's not subtle.

That's quite an identity shift.

And we're going to actually get into identity
when we look at some of the evolutionary steps later on.

But I wanted to slow down on this because I thought,
that's not a minor change.

That's a whole different perspective shift.

And I think we need to really make note of that

because I think it colors a lot of what's to follow.

What makes you say that we're a referee in this situation?

Discrepancy.

There's a difference between what the world is telling us
and then how we rule on that.

So think of any game where you get into an argument with the refs.

Are they wrong?

Are you wrong?

It doesn't really matter.

The point is that there's a difference.

That's when you said differential graphing is maybe one of the ways
to be able to make that explicit.

But that's not how it's typically framed out
when we're talking about Bayesian or Markovian stuff.

So I think this is a big move.

I know it might seem like a tiny one,
but I think as we go, again, as we go deeper into this paper,
it affirms a lot of the stuff that I actually saw
when you're trying to go out there and forage and figure.

So I'm excited to keep going.

Yeah, it could be like the observation.

The baseball player is running towards the base
and the coach is observing
and there's this ongoing prediction and observation
with no discrepancy

because part of the generative model
includes the person moving through time.

And then, given the observation and the generative model,
the coach predicts slash expects and prefers,
which we'll come to again later with the three Ps,
that the baseball player is safe.

You rarely see super animated refusals
when a call has gone towards somebody,
but it's when it has gone against,
it's violated their fitness,
that there's a discrepancy
with what the referee has called
and what the interested coach has called.

And that is going to lead to some consequences.

And again, we're still not at the Bayesian graph level,
but that's going to be the next figure.

Stephen?

This can then build on the idea of what's cognitive.

What can you have a perspective on?

So I think what you're saying there

with this referee position is,

as well as the kind of swarming dynamics

and the kind of inactive processes,

much of which is beyond our ability

to sense or integrate,

what is it once it starts to hit?

What's the big bucket at the kind of cognition level

that still isn't too much of an inflation?

So in this case here, discrepancy,

there is a sense of being able

to distinguish perception and action at some level

and find a discrepancy,

which in some ways,

I would imagine,

covers a lot of what cognition would require

to be thought of in a cognition way.

Other types of action,

for instance,

how we heal or grow,

maybe action and perception won't be so separated,

but this is trying to come at the idea of cognition.

So maybe that also informs this process.

Cool.

So section two,

again,

was just about the two concepts

of predictive regulation and control.

That's what we saw here.

Predictive

and control.

So it's just conceptually laid out.

In figure two,
they formalized brain design
as structured learning
in generative models.
So here we have a different figure.
We still have
the cognitive entity
and the world.
We add in
one layer
of
absolutely essential
act-inv terms,
which is
generative model,
generative process.
The generative model
is the cognitive entity's
model of.
The generative process
is the underlying phenomena
that gives rise to observations.
It's the difference between
the cognitive model of vision,
a generative model of vision,
and the generative process
of visual input,
which is like
photons and the sun
and all of that.
So these are
very different.
They're complementary.
But just so that we're
really clear going forward
that we're going to be using those
in their specific sense,
not using them like,

oh, well,
it's a generative model
because this is a model
of cognition
that makes me excited
and think of ideas.
That's not how we're using
generative model here.
So just to be clear on that.
Now,
we see action
as changing the world
represented by you,
and the cognition
involves partially
action selection,
policy selection,
planning as inference,
but not going to
all those details yet,
which can have some influence
on actual,
unobserved hidden states
of the world,
X star.
X is the cognitive model
of that hidden state
of the world
that is being inferred,
and Y is the observation
that then feeds back
into the cognitive model.
So that's what these nodes mean.
It's about partitioning
the cognitive model
from the generative process,
separating the generative model
from the generative process,

and we're starting to see
the traditional blanket form
with observations
having incoming statistical
dependencies
dependencies,
and actions
having outgoing
statistical dependencies.

What else do either of you
see in this model?

Yadin?

Yadin?

Yadin?

Yadin?

Yadin?

So,

let's go back to
being a referee
for a second.

The assumption
is that you're already attending,
and most of these models
that we have taken up
in the past,
especially in 37,
what they try to focus on
is getting from A to B.

What this is essentially saying
is let's add something
to that translation.

Right?

From A to B.

Whatever we're trying
to incorporate
in this representation,
let's maybe look
at the interpretation

part of it now
and the recitation part of it.
So, if I'm the umpire,
and we can all be umpires,
we don't have to do that
just in a baseball game,
this is incorporating now
a certain critical process.
I was attending,
and I saw this,
but then there were
60,000 other people
also paying attention,
and they saw that.
That's where I think
this is getting
really interesting now.
This is actually
bringing it back
to sort of,
I still think
it's an instrumental piece,
but I think now
we're going to incorporate
some of the reality.
And is,
were the 60,000 wrong?
Or is the one person
who yelled out,
safe,
wrong?
That's what I see in this,
because that you
is definitely embedded
in the generative process,
not the generative model.
Okay, thanks, Dean.
Stephen.

So the question
that comes into mind
is where the body
is in all of this.
And I sense that
the generative process
is the bigger
process with the world.
The generative model
gives a way
to access those
priors
and to give away
the hidden states.
And the cognitive model,
generally speaking,
cognition is thought
of more in terms
of deductive
and inductive
reasoning and logic.
And it could be
that there's elements
of the abductive
that could be held
within the kind of body.
And I suppose
there's a question there
to how and where that is.
I don't think anyone
quite knows the answer
to that,
but that's my thought.
So let's remember
that this partitioning
is specific
to a given instantiation
of model-based science,

just like Majid
was talking about.
So we're not going around
and assigning aspects
or phenomena
of the world
into either generative model
or generative process.
So where does the body fit in?
Well,
with respect to a model
of the body
being a structurally real thing
that gives rise
to observations,
it's a generative process.
Looking at the coin
from the other side
and thinking of the body
as a generative model
of its niche,
doing inference
on certain things
or acting as if,
in that sense,
it's generative model.
So different kinds
of entities
are not going to be
just assigned simply
to one side
or the other.
It's going to come down
to what is specifically
being discussed.
there's a few other
not complexifiers,
but first off,

just note that the notation
here is not the same
that's used elsewhere.
So we will move
towards better
and cleaner
or reformatable notation,
but like X star
as an external state
and X as an internal state
might be clearer
for some people.
It also carries
a little bit
of a baggage
that the hidden state
is exactly
what is being inferred
about the external world.
Like there's a temperature
parameter in the brain
and then there's
a temperature parameter
outside in the world.
As we explored
in the representations paper,
it doesn't necessarily
have to be that way.
There could be
a hidden state
internally
having to do
with movement
left or right
and then there's
a temperature variable
outside
and then the model,

the generative model
of the cognizer
is about movement
conditioned on
temperature observations,
but not necessarily
simply a thermometer
being instantiated
in the head.

So it's like
good to look
at this graphically
and think about
what is being connected
to what
without worrying
too much about
all of these
side
questions,
but this is
what they're
setting up
as the basis
of their
further discussion,
which is
it's about
prediction
and control
and we can
use Bayesian
graphical approaches
to represent that.
Stephen?
And that
inferred state,
as you mentioned,

it's in some ways
that's always
slightly hidden
from us.

Like how is that,
what does that mean
to be an inferred state
in terms of
is that the cognition
that's coming
out of that
in some sort
of deductive
or can it be
an effective
sense of
how well
something's going?

So again,
it's not
directly asked
here,
but
accessing that
is one of the
big problems
that happens.

So how do you
access
what has been
inferred
when it's
not necessarily
something you can
access through
sort of direct
reporting from
a subject matter

or from
a participant?

Right.

Like another
example of that
might be somebody
who has skilled
action with
respect to
investing,
but they may
not be able
to give an
estimate for
a number
that's a
certain,
you know,
asset is expected
to be at
because it's not
like they're doing
the asset price
prediction and
then doing a
strategy.

The cognitive
model may have
a very different
structure.

So we explored
it in the
representations paper,
but like there
would be ways in
which therefore
that investment
decision is not

a representation
of the actual
stock market
because it's
not the same
variable,
but then the
aboutness of
the investment
decision would
be a
representation with
respect to
what was happening
in the generative
process.

So those
are some of
the like side
avenues that
we've looked at
previously,
but they're all
in play at
once.

And so the
question is just
how to linearly
structure to
respect the
specific contributions
that are made
here and the
insight that can
be gleaned
without every
single time
pulling back to

some of these
questions.

But it's great
that we have
like specific
papers and
memes and
core terms that
we can refer
to and then
carry on with
what they
actually contribute.

Okay.

Any thoughts
on one or
two before we
get into the
structure of
the allostat
or the homeostat
first,

I guess?

Let's go.

Let's begin.

So we're
going to go
from this
black and
white,
you know,
Dorothy still
in Kansas
mode to
some predictive
motifs of
ancestral brains.

And the

three motifs,
the red,
green,
and blue
are homeostasis.

So returning
to a set
point,
allostasis,
anticipating or
approaching a
set point in the
future, which
could be a
fixed or a
changing one,
and then
implementing
behavioral
control,
not just
scalar
homeostasis
or allostasis
on an
interoceptive
variable.

Okay.

Citation

20 in the

paper,

Chance et

al,

from the

future,

March 22,

is where

to look for

more details
on the
homeostatic
formulation that
they're using
here.

But we can
see it in
terms of their
figure 3.

Okay.

So the
left side,
just for
reference,
there's
figure 2,
so that we
can remember
the structure
of perception,
cognition,
action,
impact that
the authors
are working
with here.

And we're
going to
connect this
black and
white figure
2 to
figure 3a.

This is a
graphical
model,
graphical

in both
senses,
meaning visual,
like we're
perceiving it
through computer
graphics,
and graphical
meaning like
a network,
so nodes
and edges,
because there's
computer graphics
that aren't
network
topologies.

But this
happens to be
a computer
graphic that
we're perceiving
visually that
also is
reflecting a
graph in
terms of nodes
and connected
edges.

It's a
generative
model for
the regulation
of a single
interoceptive
variable.

So here's
A and B

with the
homeostat.

And C
we split
out to
talk about
later.

But first
A and B.

This
generative
model includes
an interoceptive
thermoreceptor
and a
belief about
body temperature.

The prior
over X ,
which is
body temperature,
is kept
fixed,
and hence
it acts
as a
cybernetic
setpoint.

Any
discrepancy
between
the
predictive
thermoreceptor
activity given
beliefs about
 X ,
so Y

conditioned on
X,
and the
measured Y
is registered
as a
prediction error
that is
cancelled out
by an
autonomic
response U,
for example,
a thermoregulatory
response.
So hidden
state,
on temperature,
beliefs about
temperature,
Y,
the thermometer,
reception,
and then
there's the
selection of
action,
so some
sort of
vasodilation
or
thermoregulatory
response.
And then
that's going
to change
the underlying
unobserved

true temperature,
but that's
not needing
to be shown.

So any
comments on
that first
part?

We're looking
just at the
top half
of 3A
and connecting
X,
Y,
to
U,
action.

Can you
still hear
me?

Yep.

Can you
take the
cursor now
and just
reinforce the
feedback and
the feed
forward part
of this?

Because the
authors spent
a bit,

I don't
want to
sort of
unpack like

they have
at the
figure 3
unpacking
that we
have on
the bottom
of the
slide.
Can you
just run
the cursor
over all
the examples
of feedback
and feed
forward
going on
concurrently?
because I
think
that's
really
important to
see that
it's
happening
in both
directions
at once.
Let's
label
everything
and then
definitely
can do
that.
Perfect.

Thank
you.

Okay.

So
light blue
is
action.

These are
subtracted.

So red
circles
represent
expected
values
of

x.

The red
circles
which are
used to
make predictions
about y.

These are
subtracted
to form
a prediction
error.

because those
lines with
the arrows
on the end
of them
are just
dependencies.

They don't
really show
both the
feedback

and the
feed
forward.
Yes.
Agreed.
Like having
some
rounded
edges
and other
directed
edges.
Let's see
whether it's
whether they're
like kind
or not.
Let's
start with
action.
Action
influences
the state
of the
world
which
an
edge
can be
drawn
to how
that
changes
the state
of the
thermoreceptor.
y here
was the

observation.

Yeah.

That's the
state of the
thermoreceptor
is the
observation.

Okay.

Yeah.

So action
changes the
observation,
the state
of the
thermoreceptor
which is
being
contrasted
with the
belief about
temperature.

Yes.

From there
a prediction
error
is generated.

It could be
zero if there's
no difference
or it could
be higher.

Can you
can you
just show
with a
green
some colored
line that

has an
arrow on
each end
a connection
that demonstrates
within that
diagram the
feedback and
the feed
forward at
once.

Is that
possible using
this diagram?

Let's see.

This representation

so a single
arrow that
or a single
line that has
an arrow at
both ends

is a
different color
so we've
superimposed
it over

this but

that it

shows that

there's a

feedback feed

forward loop

going off

at the

same time.

Okay so

here's the

temperature
information
measurement
flowing in
to contrast
with the
beliefs and
this is it
is in general
really important
to label the
edges not just
use color
coding.
we'll let it
roll for
now.

The information
is flowing
from the
measurement
to the
belief and
then that
gives us
the prediction
error.

Thank you.
the prediction
error is
used in
the selection
of action
which then
influences
future
observations.

Now the

red circles
represent
expected values
of x .
I'm actually
not sure
what exactly
the red
bottom
larger circle
is meaning
because the
prediction
error.
Well there
has to be
some way of
representing that
discrepancy
right?
So they
needed the
second ball
on the
bottom to
show difference
between
prediction error
and expectation
I think
that's all
that's
trying to
show.
Yeah or
another
possibility
might be

that the
prior is
staying
fixed.

Correct.

Yeah.

Priors can
also
they can
be
flexible but
in this
case it's
a fixed
prior.

That's
because we're
talking about
homeostasis.

Then
the
observations
are
diverging
from
the
prior
and the
posterior
is kind
of like
the
realized
perception
which is
a compromise
between the
sensory data

coming in
and the
prior.
And so
yes there's
a lot of
degrees of
freedom
depending on
how parameters
are weighted
this green
line might
approximate
the red
a lot more
sharply.
We'd call
that a
weaker
prior
because
sensory
data
updates
to be
more
like
it
or
it
could
be
the
case
that
having
a lot

of
observations
different
from
your
prior
don't
change
it.

That's
a strong
prior
where
sensory
data
do not
change it
as much.

But then
here
is the
prediction
error
in
relationship
to
x.

I don't
know.

We can
look at
the paper
but is
mu
shown
in an
equation?
I don't

have it
copied out
if it
is.

Let me
look here
while you're
doing that.

Yeah.

But that
is the
perhaps
here.

Okay.

Stephen
anything?

Yeah.

I'm just
noting how
the
effect

the
belief

on
temperature

it
flows
through

the
thermoseptor
down

into
the
to
get
the
expectation
and

prediction

error.

So

it's

like

you've

shown

there

there

is

a

dynamic

going

down

from

the

belief

through

to

prediction

error

being

mediated.

It's

like

the

thermoseptor

is

kind

of

like

a

mediator

between

belief

and

prediction

error

on what
to do
for
action.
And
another
point
I
the
reason
why
you
can't
find
the
view
is
because
it's
not
actually
pointed
to
in
the
description.
It's not
in the
caption.
I
remember
reading
and
looking
and
looking
and
looking

and
not
being
able
to
find
it
and
then
looping
back
up
up
paper
to
go
okay
so
how
about
feedback
feed
forward.
now
there
are
cases
where
mu
is
used
to
describe
internal
states.
That
may be
implicitly

how it's
being used
but it's
super important
that all
variables are
defined in
a paper.

I wish
they all
had a
table
for every
single
variable and
expression
that were
used.

yeah it
would make
the dot
zeros
easier but
also it
would reduce
uncertainty.

For
example is
this epsilon
even
described?

Okay Stephen?

I mean I
suppose in a
thermostat in
some ways the
internal
states of

the
bioelectric
strip in
some ways
it's kind
of it
holds the
kind of
the way
that
the
expectations
of action
can happen
because in
some ways
it dictates
the way
that the
thermostat will
behave in
some sort
of ways even
if it's kind
of an
analog
process.
So I'm
thinking
where would
the kind
of inner
thermostat
where would
I know this
is a
homeostat so
it's a bit

more sophisticated
now but
extrapolating that
out you've
kind of got
a belief
and there's
a belief of
what something
is and then
there's an
expectation of
what you can
do about it.
For instance
my beliefs can
go bigger than
what I can do
in my actions.
I could have
beliefs about
temperature which
exceeds where
I could even
exist or where
I am able to
change it
you know
depending on
the scenario
so it's sort
of held in
both scenarios
in both
the body
and the
context
and the

kind of
the probabilities
available.
Okay so
for a
non-living
thermostat
it doesn't
have a
cognitive
belief on
temperature
but there
could be
something that's
computationally
like that
reflected by
like just a
digital prior
on temperature
and again
these aren't
cognitive
personal
affective
experienced
beliefs
Bayesian
belief
this is just
saying random
variable reflecting
in a model
variable on
temperature
so
even a

sincerely held
incorrect
psychological
belief
is not the
Bayesian
belief
they might
coincide at
times
if it were
a parameter
but
the belief
here being
the prior
it must
be adaptive
that's the
evolutionary
twist
that actually
helps resolve
a lot of
this because
otherwise
right the
design space
of all
edges by
all nodes
and then
any variable
I mean it's
just like saying
here's all the
words in the
dictionary

and so
evolution
helps
restrict
the discussion
to
cases that
actually do
manage to
achieve
adaptive
control
Dean
and this
is where
that
translation
from the
silhouette
of a
head
to a
statistical
density
is assumed
that the
person who's
following along
with this
just sees
that
but you can
also see
how easy
it is to
slip
into the
idea that

oh wait a
second we've
gone from
the physical
space
and we just
held on to
the physical
space when
really now
we're talking
about a
statistical
density
space
and again
if you're
not really
really careful
you can see
how people
can carry
forward
something
but the
actual
thing that
they're
talking about
has changed
that's why
the word
discrepancy
was such a
big deal
to me
because I
normally

gloss over
things but
this time
it actually
I went
oh okay
so there's
going to be
some
moments here
where we're
actually talking
about different
things
even though
in the
continuum
we kind
of think
that they're
talking about
the same
thing
no we're
not
yep it's
the travails
of realism
and instrumentalism
for biological
active inference
episode 55
because
are these
terms are
they an
example
or is this

an example
of a model
being used
to discuss
a real
system
but
this suffice
to say
is the
architecture
of the
homeostat
it
undertakes
action
to reduce
discrepancy
relative to
a prior
held belief
fixed in
this case
about what
temperatures
are
expected
slash
preferred
the dialectic
of the first
two Ps
from
live stream
number 37
this is
an expectation
and a

preference
because
expectations
having to do
with survival
are as good
as preferences
for survival
over evolutionary
time
okay
this
is going
to be
contrasted
with
figure
3b
which is
the
allostat
so we
see the
same
stack
of
 x_1
 y_1
light blue
dark blue
red
except
there's now
a second
column
next to it
and there's
some cross

connectivity
so again
the x's
are going
to be
beliefs
about
so
priors
on
and then
these are
observations
and so
they're saying
this generative
model extends
the homeostat
by including
a second
set of
exteroceptive
variables
that correspond
to light
intensity
y²
observations
of light
and beliefs
about
sunrise
so like
beliefs
about the
generative
process
so generative

model
of the
generative
process
furthermore
the model
includes a
predictive
relationship
between
sunrise
 x_2
and body
temperature
 y
so
again
i wish
we could
label
every
edge
because
the edges
are meaning
different
things
even at
different
times
like
this is a
predictive
relationship
so it's an
anticipatory
one
but then

the one
about
light
intensity
and the
belief
about
sunrise
there
would
be
ways
to
make
that
an
anticipatory
or an
instantaneous
relationship
but the
result of
this
sketched
architecture
is that
inferring
a sunrise
which will
only happen
with high
posterior
confidence
if the
visual
observations
are
consistent

so low
prediction
error
inferring
a sunrise
not
quote
seeing
a
sunrise
that is
not
what is
happening
in the
model
inferring
a sunrise
and
finding
that
visual
observations
are
compatible
with
it
can
trigger
the
autonomic
response
the
behavior
you
of
thermoregulation
in an

anticipatory
manner
that is
before
sunlight
actually
increases
body
temperature
well
how
would
the
parameter
be
set
that
way
because
if
that
were
adaptive
then
other
parameter
combinations
have been
weeded out
already
by evolution
so
that is
where we
tuck
the thread
back into
the ball

of yarn
which is
the parameter
combinations
that are
non-adaptive
die
they dissipate
they fail to
exist
they're not
going to be
measured as
things
in the future
empirically
however you
want to
take it
so
like
the quote
like
how does
it work
it's the
same as
it was
50 years
ago
which is
it has
to do
with
survival
of the
persistent
Stephen

and
these
these
graphs
what they
are used
for as
well
is
they
do
show
beliefs
tied
to
the
observation
space
that the
space
available
around
the
observation
you know
whatever the
sensorum
that is
available
that's
what gives
the beliefs
its scope
and then
the
expectations
are
tied in

here
more clearly
to action
so the
errors
the errors
of the
what's
important
the
actual
dynamic
that's
driving
action
to change
something
is
coming out
of the
prediction
area
error
that's
feeding
into
action
I think
that's
quite
important
as
because
this is
like the
proto
animal
this is

like the
proto
piece
here
that ties
into
a lot
of how
we think
about
knowledge
and meaning
and I'm
not saying
that I'm
just
extrapolating
a lot
but I
think it
does
show
how
beliefs
are tied
to
the
type
of
sensorium
that is
being used
to shape
the observation
space
and I think
that's
quite useful

yeah
and these
are just
yeah
Dean
first
go ahead
I just
want to
get your
gut
both
you guys
opinions
about
that
orange
barbell
at the
bottom
of the
diagram
because
talking
identity
but we're
also still
talking about
dependency
we're talking
about
duplication
we're talking
about
anticipation
so
that's a lot
of stuff

packed in
one orange
barbell
what do
you think
yeah
I'm going
back to
the full
caption
the red
circles
represent
the expected
values
of x
so
if we're
interpreting
these
as the
posteriors
on x
because
the
expectations
the priors
are the
top
blue ones
so
it wouldn't
have to be
the red
circles
and they
don't
mention

the word
orange
note
the
lateral
modulatory
connections in
the
allostatic
network
c24
for
details
and
24
is
the
graphical
brain
frist
and
par
and
dev
rise
so
yeah
again
this is
just a
sketch
model
they're
not
using
it
to
fit

any
data
or
even
simulate
any
data
and
it
does
relate
to
what
steven
said
about
the
inter
sensory
or
intermodal
inference
which is
one could
imagine
a cognitive
model
where
if
sunrise
if there's
noises
that are
associated
with
sunrise
then
beliefs

about
sunrise
can be
a variable
with edges
coming from
different
sensory
modalities
so it
gives us
a separation
of
the
organs
of
sense
and
internal
cognitive
modeling
steven
yeah
it gives
a sense
of when
would
something
be
important
to act
upon
so there's
many many
things that
I could
believe
I'm seeing

or being
perceived
in terms
of this
is
sunrise
or this
is
you know
this is
the type
of light
coming in
this is
the nature
of the
light
this is
where
the light
is coming
from
there's
all sorts
of stuff
the stuff
that's
really
filtering
down
is
what
is
important
important
and this
is where
the affordance

is coming
what affords
me to
make some
action
if it
was
expanding
out
there can
be lots
of things
I notice
like I
might notice
that there's
a line
there and I
may notice
and believe
it's orange
but I may
not act
upon it
until I
take my
attention
to it
and think
it's
useful
as there's
many things
going on
on the
page
you know
so when

we're
we have
beliefs
but then
there's
what's
being
acted
upon
out
of
all
of
that
to
then
create
some
sort
of
change
so I
think
that
is
also
quite
useful
in
this
architecture
cool
can I
add to
this
because I
think
it's

really
important
so we're
talking
about
in the
context
of
a
sun
coming
up
but
let's
let's
see
whether
it's
still
is
true
and I
think
it
is
just
to
prove
a
point
whether
that
orange
barbell
still
makes
sense
if

we're
talking
about
Daniel
anticipating
61 year
old
Daniel
and
Stephen
anticipating
61 year
old
Stephen
and
Dean
anticipating
61 year
old
Dean
now I'm
much
closer
to
61
than
the
two
of
you
but
do I
need
the
Monte
Carlo
do I
need

the
play
out
to
be
able
to
make
that
anticipation
as
long
as
I
have
the
identity
and
the
parallelism
and
the
backwards
and
forwards
looking
dependency
that
we
see
between
X1
and
X2
and
Y1
and
Y2

that's
what
I
think
makes
this
a
very
interesting
way
of
being
able
to
build
on
the
previous
homeostatic
example
thanks
Stephen
yeah
I think
also
this
speaks
to
traditionally
we
think
about
the
meaning
what
does
this
mean

what's
all the
meaning
out there
as you
mentioned
all the
data
the
stuff
that
it
could
mean
to
relate
all
these
variables
but
ultimately
and
the
bit
that
is
actually
what
active
inference
gives us
is
that
barbell
is
where
the
meaning

comes
in
it's
like
the
way
to
know
how
being
61
is
for you
Dean
is
for
you
to
imagine
what
it
would
be
like
to
make
choices
and
to
act
in
the
world
as
a
61
year
old

Dean
not
necessarily
to
go
out
and
look
at
all
the
data
so to
speak
all
the
beliefs
and
the
things
we
think
about
the
world
but
to
actually
bring
that
in
and
of
course
you've
got
a
better

chance
to
do
that
because
you're
closer
to
being
that
kind
of
Dean
so
I
think
that's
quite
interesting
that
barbell
at
the
bottom
in
terms
of
pairing
together
the
kind
of
beliefs
about
what
actions
and
prediction

areas
are
available
and
it's
more
into
that
meaningful
action
meaningfulness
realm
rather than
what does
it mean
what's
the data
tell us
so
yeah
we might
we might
get there
and I'm not
pushing back
on what
you're saying
Stephen
we might
get there
but I
think
for now
what
we're
saying
what we're
saying

is
is that
in order
to
get
there
we
have
to
have
more
than
more
than
a
single
stack
of
dependencies
we
have
to
have
a
double
stack
in order
to be
able
to
extend
the
homeostatic
nature
of
how we've
evolved
to where

we are
I don't
know
again
I'm not
pushing
back
on you
I'm
just
saying
I don't
want
to
jump
that
firing
pistol
just
yet
because
I
think
that
orange
barbell
is
going
to
turn
into
a
green
arrow
and
then
all
hell

is
going
to
break
loose
anyway
so
we
should
carry
on
I
don't
want
to
take
Steven's
stuff
away
but
I
want
I
think
I
want
to
park
it
because
we're
going
to
go
somewhere
in a
minute
I'm

hesitant
to
ascribe
too
much
specificity
to
something
that
wasn't
labeled
in the
caption
doesn't
have
a
clear
labeling
in
the
figure
either
but
I
think
it
is
already
demonstrated
that
using
graphical
frameworks
and
partitionings
like we
have
in

active

inference

we

can

start

to

approach

some

of

these

questions

like

how

would

different

kinds

of

variables

be

connected

how

does

that

relate

to

future

inference

on

action

counterfactuals

etc

okay

so

these

sections

4

5

6

7

were

just

laying

out

the

relationship

between

an

evolutionary

or

physiological

function

like

homeostasis

allostasis

or simple

behavioral

control

models

of

things

that

are

core

evolutionary

features

and

functions

connecting

them

to

the

kinds

of

graphical

representations

which

are
like
a
little
bit
of
a
hybrid
of
a
Bayesian
graph
and
a
factor
graph
because
there's
some
kind
of
computations
being
implied
here
whereas
in
a
Bayesian
graph
the
edges
only
reflect
statistical
dependencies
whereas
we're

bringing
in
a
little
bit
more
of
like
a
nuanced
type
relationship
that
one
could
imagine
could
be
unpacked
a
bit
more
were
it
to
be
specified
but
it's
just
to
show
how
different
graphical
models
relate
to

different
functions
architectural
underpinnings
whether you
think this
is the
actual
architecture
or
whether
it's just
an
instrumental
architecture
like a
model
structure
on
a given
phenomena
how do
model
architectures
relate to
evolutionary
functions
where
is
evolutionary
time
in
this
model
we
started
out
by

asking
how
do
cognitive
!
phenotypes
arise
specifically
over
evolutionary
but also
over
developmental
time
and that's
where we
get to
one of
the
main
pieces
and
contributions
of the
paper
which is
figure
four
the five
main
dimensions
of
elaboration
of
generative
models
introduced
in this

paper
so
there's
other
kinds
of
changes
and ways
that this
structure
could
evolve
but they're
going to
focus
on these
five
there's
starting
with the
homeostat
does it
have to be
our
starting
point
maybe
not
but
starting
with the
homeostat
there's
the
I
operation
which is
identity

unchanged

there's

I plus

I

which is

just a

parallel

isolated

duplication

that's

like

from one

photoreceptor

to going

to two

photoreceptors

so to

speak

there's

the

allostat

which is

a

duplication

as well

as a

cross

linking

so

these

are

just

sketches

one

could

also

probably

write

that
one
as
like
a
duplication
followed
by a
linking
and
there's
no
like
change
in
linking
for
example
here
so
this
is
like
a
few
of
a
taxonomy
of
operations
there's
increases
in
temporal
depth
within
a level
of the

hierarchy
so
looking
one
more
unit
further
in
time
at
a
given
time
scale
and
then
there's
hierarchical
nesting
with
an
h
and
the
figure
is
shown
like
this
because
these
are
like
the
things
that
can
happen

to
the
homeostat
and
then
they
can
have
a
second
round
and
so
on
and
so
on
there's
so
many
other
ones
that
could
happen
like
where
is
reduction
where's
loss
where's
deduplication
where's
the
uncoupling
of
the

allostat
reduction
of
temporal
depth
reduction
of
hierarchy
so
there's
a
broad
space
but
one
of
the
main
contributions
of
this
paper
is
to
connect
functionally
oriented
graphical
models
of
homeostatic
and
physiological
function
to
the
operations
that

result
in the
elaboration
of
simpler
model
architectures
into
different
architectures
steen
this
is
an
interesting
part
i
know
stephen
wants
to
say
something
too
but
really
quickly
there's
no
plus
minus
multiply
and
divide
there's
an
operation
but

there
there's
no
symbolism
for
that
and
again
that
keeps
it
safe
on
a
statistical
level
i
think
that's
a
nice
tell
there
i
want
to
again
for
somebody
who's
just
sort
of
looking
at
this
for
the

first
time
pointing
that
out
it's
not
there
and
because
it's
not
present
that
tells
us
something
to
your
point
about
maybe
how do
we get
deduplication
yes
it's
well
i
plus
i
it's
a
suggestive
use
of the
addition
operator

but
we're
not
adding
these
just
like
integers
so
yes
they're
kind
of
like
categorical
operations
that
constitute
this
evolutionary
algebra
okay
steven
anything
on
figure
four
yeah
i was
just
saying
that
i
was
saying
they've
set
a

kind
of
a
paradigm
from
the
homeostat
and
they've
taken
that
through
and
they've
shown
that
it
plausibly
can
carry
through
to
cognition
and
then
like
you
say
there
can
be
other
ways
but
what
they've
now
given

is
they've
given
some
legs
another
way to
think
of the
legs
that
can
be
attached
to
that
paradigm
because
this
modeling
approach
while
they
could
have
chosen
a lot
others
by
taking
it
there
are
certain
things
which
are
hard

baked
into
that
early
stage
choice
making
which
then
makes
other
paradigms
wouldn't
fit
with
it
right
you
know
you
certain
certain
even
if you
go
instrumental
or
realist
you know
your
instrumental
models
your
realist
models
don't
make
sense

unless
you
with
this
unless
you
take
in
some
of
the
paradigms
that
this
structure
creates
yeah
agreed
thanks
just
maybe
think
like
is
this
functional
model
realism
like
the
function
of
the
model
is
what
is
being

tracked
through
time
and
so
that
there's
many
other
alterations
that
could
occur
and
then
there's
the
even
more
granular
like
parametric
changes
or
changes
in
connectivity
or
edge
types
and
then
I
think
as
you
said
Dean

when
all
hell
breaks
loose
so
then
it
just
what's
interesting
here
let's
look at
the
allostat
in a
little
bit
closer
detail
okay
so
previously
we
looked
at
the
allostat
and
we
were
like
okay
there's
an
orange
bar

now
the
allostat
has
a
green
bar
it's
a
green
arrow
and
it's
a
unidirectional
one
this
one
the
dependency
um
the
allostat
has
the
green
bar
oh
okay
well
those
are
my
bad
eyes
because
I
saw

I
saw
the
little
arrow
from
the
blue
actually
turning
I
didn't
actually
see
that
there
was
two
circles
thank
you
there's
my
60
year
old
eyes
for
you
yeah
um
okay
cool
um
and so
each
of
these

are
in
quite
a
different
domain
um
or
at
least
like
a
different
aspect
of
cognition
and
these
are
the
transitions
that
it
can
engage
in
so
like
duplication
retains
the
exact
same
function
but
it's
a
very

important
step
and
that's
a little
bit
what
I
explored
in
the
dot
zero
with
genetic
duplications
like
on the
left
side
here
of the
screen
having
a
region
of
the
genome
duplicate
yes
one of
the
duplicate
copies
can
just
degenerate

that
may
be
like
opening
up
space
or
creating
motifs
for
other
kinds
of
later
evolutionary
steps
so
even
this
is
not
like
a
failure
it's
just
a
change
there's
neo
functionalization
where
then
those
two
models
can

start
to
subspecify
like
if
you
only
have
one
photoreceptor
it has
to be
doing
what
it's
doing
but
duplicating
it
then
allows
one
to focus
on a
different
wavelength
than the
other
for
example
and
then
there's
sub
functionalization
which
is
where

initially
a
composite
function
AB
becomes
two
like
enzymes
start
to
subspecialize
on
one
of
the
two
functions
of
the
ancestral
so
this
is
kind
of
like
a
role
duplication
okay
so
duplication
is an
important
event
then
there's

the
allostat
and
multiple
things
happen
with
A
but
we're
seeing
it
as
there
is
a
duplication
occurring
and
there's
also
cross
linking
!
occurring
between
beliefs
in
one
modality
or
beliefs
about
one
type
of
thing
influencing

observations

or

about

another

kind

of

thing

as

well

as

this

lateral

modulatory

green

orange

bar

we

have

and then

we have

two

kinds

of

expansions

into

temporal

depth

a

sort

of

local

expansion

with

a

T

operation

just

pushing

the
model
one
step
deeper
and
then
the
H
hierarchical
nesting
operation
which
brings
it to
another
timescale
of
analysis
Dean
and
this
is
where
I
really
struggled
and
that
is
in
when
they
were
at
Allostat
it
wasn't

just
comparative

I
felt
like
you
had
to
build
in
time
especially
if
we're
talking
about
identity
but
then
it
was
like
no
we're
just
going
to
leave
it
comparative
and
then
we're
going
to
introduce
how

it
so
is
temporal
only
going
forward
in
time
I
don't
mean
all
hell
broke
loose
but
I
did
ask
this
question
you're
right
here
there's
the
triggering
in an
anticipatory
manner
and
so
in
one
view
it's
like

but
this
is
a
single
time
step
model
so
how
could
that
occur
because
it
doesn't
have
a
temporal
depth
on
the
other
hand
again
like
kind
of
those
two
senses
of
representation
in the
structural
sense
if
the

model
is
a
single
time
step
model
it
cannot
have
a
representation
through
deep
time
ergo
it
cannot
be
anticipatory
functionally
the
other
side
of
the
coin
would
be
if
it's
enough
at
this
time
step
such
that

beliefs
about
sunrise
trigger
and
response
the
model
can
be
functionally
anticipatory
without
a temporal
depth
representation
right
which
is
why
the
representation
discussion
was
very
important
leading
up
to
this
one
because
we're
all over
those
eight
quadrants
here

because
in
evolution
and this
is something
that like
cognitive
psych
people
talk
about
and
is
like
our
cognition
being
shaped
towards
effective
which is
to say
productive
slash
reproductive
success
not
towards
like
ultimate
truth
discovery
and
should
we
be
too
surprised

that
our
instrumentation
which
has
had
a
certain
objective
function
is
then
susceptible
to
being
convinced
about
things
that
are
not
true
well
I
don't
socialize
with
those
people
so
I
don't
know
I
just
knew
that
when

I
was
looking
at
it
I
was
thinking
of
in
real
practical
terms
what
is
built
into
this
in
order
for
it
to
still
make
sense
and
that
wasn't
necessarily
parsed
out
in
the
diagram
you
kind
of

had
to
fill
that
in
through
your
own
interpretation
of
what
was
going
on
so
thus
back
to
discrepancy
that
discrepancy
thing
is going
to come
up
again
and
again
and
I
really
glad
that
they
inserted
it
at
the

beginning

thanks

Steven

I

think

it's

useful

here

to

hear

this

temporal

depth

and

hierarchical

depth

being

sort

of

spelt

out

so

because

I

think

sometimes

it's

a little

confusing

in

other

papers

quite

where

it

fits

so

if

temporal
depth
and
this
again
might
be
where
some
of
ontology
work
can
be
useful
just
to
see
if
that
can
be
consistent
but
if
temporal
depth
is
you've
got
variables
for
past
present
and
future
states
but

they're
operating
maybe
within
the same
kind
of
dynamics
and
then
if
you've
got
different
temporal
rates
operating
simultaneously
then
you get
that
hierarchical
depth
and
I
thought
in
the
past
I
sometimes
thought
temporal
depth
would
have
covered
that

more
hierarchical
depth
so
you know
this is
interesting
to think
about
whether
this
might
be
actually
the
way
that
that
differentiation
is
best
made
differentiation
is probably
best made
with reference
to a
specific
model
and this
gives us
the
design
language
and the
grammar
and the
motifs

so
if we
want to
predict
100 years
in the
future
should we
have
a
decade
model
with
10
year
models
that's
a
times
depth
of
10
decades
and
then
a
nested
model
with
a
depth
of
10
years
or
will
we
have

a
one
layer
model
with
100
they're
very
different
and
it's
not to
say
that
one
is
more
accurate
or
less
and
depending
on
how
the
exact
situation
is
set
up
maybe
the
computational
requirements
of
one
or
higher

or
different
than
the
other
but
that's
the
discussion
to
have
do
we
want
to
have
a
B
transition
matrix
let's
look
at
a
nested
model
with
temporal
depth
so
here
we
have
on the
top
level
of
the

model
S
state
2
this
is
temporal
depth
happening
at the
upper
level
of
the
model
three
discrete
time
points
in
an
upper
level
that
is
temporal
depth
there's
also
nesting
within
each
time
step
there's
a
cognitive
rollout

on
three
time
steps
this
is
one
type
of
nested
model
but
there's
other
nestings
that
can
exist
so
here
we
see
a
depth
of
two
and
then
at
each
level
of
the
nesting
there's
three
temporal
depth

but
they're
totally
different
in
architecture
and
in
function
so
it
is
the
interesting
discussion
like
especially
when we
have to
do
multi-scale
prediction
like
Fermi
estimation
so
if the
prediction
is within
one
time
scale
there
isn't
necessarily
a need
to
nest

but
once
we
get
into
larger
time
scales
or
where
there's
recurrence
over
multiple
time
scales
it
makes
sense
to
have
a
nesting
model
like
if
we
wanted
to
predict
someone's
activity
over
the
next
seven
days
it

might
make
sense
to
have
a
day
model
with
a
nested
model
inside
of
that
because
then
within
each
day
there
could
be
parameter
setting
versus
just
trying
to
fit
one
time
series
and
find
the
parameters
that

make
that
one
time
series
oscillate
in
a
circadian
way
instead
we
just
need
a
day
model
with
a
very
specific
kind
of
simple
transition
and
then
a
transition
at
a
nested
level
with
a
depth
that's
measured

at
the
hourly
or
the
minute
timescale
rather
than
the
days
and
that
becomes
especially
important
when
doing
family
based
model
fitting
like
variational
inference
so
we're
not
just
like
getting
all
the
data
points
and
fitting
a

spline
through
them
but
we
actually
need
to
right
off
a
cliff
just
like
we
talked
about
with
axel's
bacterium
so
if
we
want
to
do
free
energy
minimization
and
model
selection
on
model
structures
we
can
find

the
best
model
relatively
easily
given
the
families
that
we've
specified
but
there's
no
guarantees
once
we
step
outside
of
that
spotlight
that
we're
even
in
the
right
category
structurally
and
so
that's
why
humans
having
these
design

motifs
in
mind
and
many
many
many
more
helps
us
not
get
into
what
seems
to be
a
global
optimization
based
upon
free
energy
minimization
but
actually
is a
very
relatively
local
optimization
based
upon
unimaginative
model
structure
learning
Stephen

yeah
thanks
that's
really
helpful
and
it
also
ties
into
something
I've
been
really
thinking
about
in
terms
of
the
model
in
itself
is
these
scales
these
steps
you know
you're
talking
about
the
multi
year
decade
when
we're

modeling
it's
like
what
do
we
have
access
to
so
we
tend
to
ascribe
that
in
terms
of
what
can
we
externalize
and
what
can
we
use
to
put
something
into
a
model
if
it's
instrumental
and
then

of course
there's
a
question
of
well
what
is
the
realist
perspective
on
that
and
what
is
there
that's
within
our
conscious
awareness
of
being
in
the
world
something
that
can
be
reported
as
an
event
a
story
and

what
time
scales
are
present
there
when
you
get
into
100
years
it's
beyond
that
so
then
you're
saying
okay
what
are
you
going
to
extrapolate
what
sort
of
temporal
structure
but
on
the
other
scale
if
I'm

asking
what
do
I
think
I
expect
you
to
do
next
or
Dean
to
do
next
and
it
might
seem
intuitive
I
can
have
a
whole
series
of
temporal
and
nested
scales
going
on
which
is
not
even

available
to
me
in
a
cognitive
conscious
way
I
might
have
a
sense
of
a
feel
for
what
I
think
is
going
on
but
a
lot
of
my
predictive
processing
will
be
happening
at
rates
unavailable
to
my

conscious
awareness
so
that
also
ties
in
I
suppose
I
am
not
saying
it
adds
to
the
challenge
I
suppose
but
it
is
the
challenge
of
how
much
ends
up
being
what
we
can
get
a
measure
on

to
put
into
our
model
as
much
as
what
it
is
and
that's
the
same
challenge
in
a
way
that
comes
even
from
the
realist
perspective
at
some
level
as
an
organism
what
is
it
that
I
even

can
access
in
some
realistic
way
not
saying
we
can
model
it
or
know
it
could
be
put
into
something
like
this
sorry
Dean
you're
going
to
say
something
well
all I
was
going
to
add
is
that
I

mean
so
the
paper
is
talking
about
how
do
we
use
certain
math
to be
able
to
understand
maybe
the
evolution
of
cognition
right
that's
essentially
what
the
paper
is
trying
to
point
us
in
that
direction
so
again

in
order
to
make
this
accessible
in
order
to
sort
of
lower
the
barrier
for
people
who
actually
want
to
have
this
make
sense
because
they
don't
necessarily
spend
sums
and sums
of
hours
on
trying
to
figure
this

stuff
out
like
the
authors
or
maybe
even
us
I
want
to
point
out
something
that
I
think
is
obvious
to
us
but
maybe
not
so
obvious
to
people
who
are
looking
at
this
and
going
I
have

no
idea
what
they're
saying
I
think
first
of
all
is
that
there's
a
chronology
which
we
keep
going
back
to
not
just
at
the
temporal
depth
level
so
that's
relativity
math
there's
an
evolutionary
aspect
of
this

so
that's
algebraic
math
and
there's
a
dependency
so
there's
a
statistical
math
do
we
have
to
be
poly
math
in
order
to
be
able
to
feel
our
way
through
math
first
Carl
Friston
said
you
don't
have

to
be
poly
math
but
you
better
be
at
least
somewhat
comfortable
feeling
comfortable
with
the
math
because
there's
no
one
math
that's
going
to
get
you
here
it's
a
blending
of
all
of
them
and
then
it's

how
you
feel
about
that
that's
probably
going
to
be
the
thing
that
lowers
the
impediment
to
really
being
able
to
use
this
now
in
real
practical
terms
going
forward
so
what's
the
next
evolutionary
step
right
it's

how
you're
able
to
turn
this
into
something
so
I
think
that's
I
think
that's
one
of
the
great
gifts
of
this
paper
if
you're
not
afraid
of
being
if
you
don't
consider
if
you
don't
identify
now

or
you
don't
duplicate
now
or
you
don't
allostat
now
as
a
polymath
maybe
maybe
that's
something
we
we owe
people
that want
to get
into
this
maybe
the
British
pronunciation
of
maths
helps
reflect
that there's
already a
plurality
of maths
and
Kirby

Erner
coming
more from
the
synergetic
side
talks a
lot about
this
actually
like how
the
discourse
around
math
as
universal
language
makes
it
sound
like
math
is
a
singular
language
when
actually
math
is
a
plural
verse
and
so
maths
reflects

that
a
little
better
better
it's
not
like
these
are
just
sort
of
slopes
on
one
mountain
and
that's
Mount
Math
and
it's
so
high
and
only
a
few
people
get
to
the
top
and
scale
every
side

this
is
just
like
we're
using
maths
yes
of all
different
kinds
Stephen
and
often
you
see
a lot
of
this
in
the
papers
that
published
around
active
inference
there's
often
a
team
of
people
a
group
of
people
where

maybe
one
member
has
that
higher
level
understanding
of
math
but
the
other
question
is
to
understand
what
would
it
mean
to
make
an
observation
now
that
could
be
a
very
the
mathematician
may
not
be
the
one

what
like
some
of
the
work
with
Ryan
Smith's
work
on
the
gut
you
know
it
was
like
well
they're
on
what's
the
gut
doing
so
they
created
some
electrode
I'm
not
quite
sure
how
they
did
it

but
they
had
some
way
of
getting
sensory
information
effectively
sensory
information
or
effectively
some
sort
of
information
to put
into
their
model
and
so
knowing
how
action
and
observations
can
flow
in
and
the
implications
of
that
I

think
is
also
kind
of
a
big
part
of
this
right
and
in
some
cases
it
may
be
a
trivial
part
because
it's
fairly
obvious
in
other
parts
that
may
be
the
biggest
barrier
to
and
then
that

may
be
the
biggest
barrier
as
well
often
in
organisms
right
how
do
you
how
do
you
even
access
plausibly
the
approximation
science
which
isn't
so
approximate
approximate
it's
just
chaos
that
it's
tractable
yeah
the
rate
limiting

step
for
impact
and
improvement
in the
real world
is unlikely
to be any
single
individual's
conceptualization
of math
on a
team
if it's
structured
appropriately
and
for those
pushing the
frontiers of
math
their
understanding
of math
is quite
literally
a rate
limiting
step
for them
or maybe
it might be
something very
mundane
like
time

availability
but
when it
comes
to
thinking
about
real
model
based
science
and
translational
applications
of
active
inference
I think
that we're
working
towards
new
ways
of
combining
skills
and having
shared
knowledge
resources
that help
make that
make sense
and using
shared
language
and
ontologies

narratives
formal documents
tools
onft
because
we're
doing
it
on
teams
and
we're
online
let's
just
in the
last
20-30
minutes
go through
the final
pieces of
the paper
so that
this .1
will have
been like
a first
sweep
initial
pheromone
deposition
and then
we can
return
and take
some
cul-de-zacs

etc
so
section
9
explores
a little
bit how
the
duplication
operation
allows
for
multiple
behaviors
to arise
and
that's
sort of
by
analogy
to
the
genomic
duplications
and
specializations
and all
those
different
routes
that can
occur
here
they
connected
duplication
a little
bit more

directly
to
the
factor
graph
models
in the
sense
that
duplicated
motifs
have
dynamics
that are
conserved
over
different
sensory
motor
domains
so
let's
just
say
that
we
just
had
a
visual
model
and
then
we
duplicated
it
so
now

we
have
two
photoreceptors
with
parallel
columns
and
now
the
photoreceptor
in the
second
one
changes
into
a
chemoreceptor
because
instead
of expressing
rhodopsin
protein
it's
expressing
an
olfactory
receptor
protein
there
can be
a
conservation
of the
dynamics
of
inference
even

when
the
observation
has
changed
but
it
was
a
slot
for
observation
and
so
we
went
from
having
like
monocular
vision
to
binocular
vision
to
monocular
vision
and
olfactory
perception
but
you
can't
get
there
in
one
jump

to
duplicate
and
transpose
it's
not
likely
to
happen
in
an
evolutionary
context
and
we
can
see
the
structure
of
these
graphical
models
as
equivalent
to being
factorized
probability
distributions
which is
to say
that the
sparse
connectivity
of
variables
means
that we

can make
parts
of the
model
that
can
be
fine
tuned
independently
in a
way
that can
be
fit
very
tractably
that's
the
factor
graph
that's
factorized
bayesian
inference
and we've
talked about
that in
other places
but just
to hear
all we
need to
say is
duplication
enables
like a
ctrl c

ctrl v
copy
paste
and then
let's edit
the other
version
but that
can now
happen
with
evolutionary
features
and functions
Dean
real quick
your hand
gestures in
this section
point to
the orthogonal
mesh
beginning
and then
point two
I'd like to
pull that
apart a
little bit
but let's
carry on
but that's
what I
took away
from this
the mesh
began
and in

the beginning

there was

a mesh

begins

we'll return

to that

for

dot

two

yeah

okay

section

10

talks

about

temporal

depth

it's

about

time

and

about

the

endowing

of

generative

models

with

temporal

depth

and

the way

that

that

supports

prospective

inference

anticipation

or
retrospective
inference
which is
like
memory
so here's
what the
operator
looks like
we have
 x sub
 τ
 x
at a
time
point
and
then
now
there's
 x
 τ
plus
one
the
next
time
point
or
 t
plus
one
could
be
now
and
then

t
could
be
the
last
moment
so
um
memory
and
like
figure
five
now
because
I don't
I still
see
figure
four
um
slide
29
okay
yeah
so the
structure
of
anticipation
and
of
memory
are
very
similar
it
just
depends

whether
one
has
the
stream
of
observations
happening
on the
left
side
and a
prospection
or the
stream
of
observations
are on
the
right
side
and in
which
case
it's
retrospection
Stephen
yeah
I mean
it sort
of ties
in
they
talk
about
what
the
police

do
it's
very
hard
for
someone
to
lie
backwards
so
you
always
get
them
to
tell
their
story
backwards
because
it's
very
hard
for
them
to
do
that
because
like
you
say
if
we're
creating
things
retrospectively
but

sort
of
playing
them
forward
so
to
speak
it
would
kind
of
tie
in
with
that
I
want to
emancipate
that
I looked
at police
in the
paper
I see
policies
is that
policies
um
it's
a great
opportunity
for my
favorite
joke
but I
won't
um

this
section
talks
about
temporal
depth
okay
from
whence
temporal
depth
various
researchers
have
speculated
that a
major
driving
force
for the
development
of deep
temporal
models
was
foraging
and
there's
a lot
of
interesting
empirical
and
conceptual
reasons
to think
about
foraging

in terms
of
temporal
and
spatial
depth
of
model
and
that's
true
in
the
vertebrates
where
they're
discussing
mainly
like
hippocampus
and
entorhinal
system
and
in
the
invertebrates
that don't
have a
hippocampus
entorhinal
system
but it's
one reason
why
comparative

Neuroscience

is so
important
because
it prevents
us from
getting
fixed
on
specific
anatomical
realizations
of given
functional
attributes
of evolutionary
systems
like
if the
story of
memory
is just
about
some
brain
region
in
humans
it may
be a
useful
model
it's not
even to
say that
it's
inaccurate
it's not
even to

say that
it's a
partial
model
it just
is a
model
of that
whereas if
we want
to understand
a given
cognitive
function
in a
broader
context
we have
to pull
back
somewhere
a little
bit
beyond
or
in
complement
to
the
anatomy
because
we need
the
empirical
anatomy
to have
anything
specific

to be
talking
about
but
it isn't
just the
case that
vertebrate
anatomy
is the
way to
do
active
inference
Steven
I think
this is
really
helpful
because
it
shows
that
the traps
you were
just
mentioning
there
that
we fall
into
is like
humans
have
the frontal
cortex
it's all
about

this new
what's
the new
parts
of the
brain
that are
there
but
we
don't
say
that
with
a
formula
one
car
we
say
well
it's
just
got
a
primitive
engine
thing
in
there
and
the
rest
of
it's
evolved
it's
like

well
that's
itself
different
like
this
and
the
same
with
the
Mark
Soames
conversation
you know
the
evolution
of
what's
often
dismissed
as
the
primitive
parts
of
the
brain
they
themselves
can
have
been
more
sophisticated
they're
doing
more

things
but
they
helped
to
be
doing
more
than
they
were
as
opposed
to
there's
something
else
that
is
doing
all
the
heavy
lifting
which
I
think
is
certainly
common
a common
misconception
I think
in psychology
anyway
Dean
can I read
quickly from

the paper
sure
the evolution
of temporally
deep
models
from simpler
models
could
be
realized
during
evolution
via
the
progressive
keyword
here
parcelization
of an
initially
undifferentiated
model
so
a
relative
sense
of
invariance
i.e.
a model
that does
not
distinguish
present
from
past
and

future
into
a
model
that
features
separate
latent
states
for
the
past
present
and
future
this
is
the
part
I
loved
a
key
drive
for
this
factorization
or
parcellation
may
have
been
the
observation
and
progressive
internalization
of the

sensory
motor
sequences
sequentially
that the
animal
creates
and experiences
while
acting
while
acting
in other
words
the
self
modeling
of
one's
own
sequential
behavior
patterns
see
53
for a
computational
example
I didn't
open up
53
but again
if we
want to
talk about
the
polymath
piece

of this
and the
fact that
our
hex
cells
have to
do
some
parcelization
it's
right there
in terms
of
sort
of
the
next
layer
on top
of this
as we
move
up
through
that
evolutionary
cycle
yeah
53
is
Stoyanov
et al
the
hippocampal
formation
as a
hierarchical

generative
model
supporting
generative
replay
and
continual
learning
there's
a ton
that could
be said
about that
from like
a
computational
and a
neuroanatomical
perspective
the
internalization
of the
sensory
motor
sequences
relates
to the
sensory
motor
detachment
that we
talked about
in the
representation
paper
so
if
some

motor
region
in the
brain
if it's
like a
marionette
and the
fingers
and then
the
motor
plant
so that
motor
region
has to
be coupled
to the
activity
of that
motor
region
like
either
in a
one
directional
way
or
maybe
even
in a
bi
directional
way
let's
just

say
so
that
system
like
as it
thinks
so it
does
and vice
versa
it
cannot
engage
in
counterfactuals
because
any
direction
that the
neural
system
turns
the
motor
system
is
just
simply
doing
that
so it
is not
able to
engage
over
evolutionary
time

in
maladaptive
action
the ones
that do
no longer
persist
it's
just
it's
always
what
ties
it
back
to
reality
and
to
the
finite
amount
of
entities
on
the
finite
spaceship
earth
like
Darwin's
famous
calculation
like
if the
number
of
elephants

slowly
reproducing
like
they'll
cover
the
earth
unless
their
population
levels
are kept
in check
and so
once
there's
sensory
motor
detachment
so
there's
some
brain
region
either
a
motor
region
or
some
supplemental
area
some
ancillary
area
that's
able to
intervene

in that
process
or somehow
play a
role
that's
detached
from
the
motor
activity
now
there
can
be
like
a
motor
planning
occurring
that
opens
up
the
affordance
of
temporal
depth
or
of
counterfactuals
all
these
other
cognitive
functions
arising
via

the
sensory
motor
detachment
and
so
foraging
is an
awesome
place
to look
at that
for a
lot
of
reasons
in
different
life
forms
and
computational
foraging
and so
like
just a
few
of the
notes
were
like
what
are
the
real
cognitive
demands
of

foraging
for
different
creatures
and
what
about
internal
foraging
like
mental
foraging
and
these
papers
are
very
good
foraging
in
mind
and
foraging
in
semantic
fields
both
very
useful
because
they have
to do
with
the way
that
they have
some
nice

maths
too
but
they
have
to do
with
how
actions
that are
spatial
can have
conserved
dynamics
structurally
just like
we explored
here
to mental
actions
and then
we see
them come
together
like with
a memory
palace
or something
like that
so
foraging
is cool
it's good
to study
11
endowing
generative
models

with
hierarchical
depth
affords
multi-scale
inference
kind of
addressed
that earlier
with like
the decades
and years
nested
temporal
modeling
is not
the same
as just
deep
temporal
modeling
and
then
and
another
way to
look at
that is
like
how many
operations
you would
need
to get
to
100
years
well

if you're
going to
do
only
temporal
depth
it's
going to
take
plus
100
or
99
time
steps
versus
if
it's
one
nesting
and then
10
on the
higher
order
than 10
on the
lower
order
it was
21
operations
and so
then
there's
shorter
sequence
of events

to achieve
a higher
performance
model
assuming
that
they're
appropriate
section
12
yeah
go ahead
just one
thing
just on
that modeling
it's also
interesting
think about
how we
think of
time
because
like
indigenous
approaches
tends
to be
well it's
a cyclical
time
right
so
things in
the future
will be
a repeat
of the

cycle
so
the times
of
a thought
of in
a cyclical
motion
of the
sort of
passing of
the day
passing of
the
and of course
we have
this
we have
this assumption
of time
stretching out
you know
so is it
that we
will
over time
is it
thought of
well what
will things
be like
in four
generations
and
and if
my world
or my
ecology

follows
certain
cyclical
patterns
and it's
fairly stable
maybe that's
quite a good
way to
think about
you know
it's like
say plant
plant the
tree now
to help
the people
in 150
years
you know
so
it sort
of comes
back to
there's
different ways
to construct
how we
think about
time
cool
so
12
looks
into
a
phylogenetic
tree

of the
evolution
of generative
model
so
what is
a phylogenetic
tree
and what's
the relationship
between
active
inference
slash
fep
and
evolution
well
we'll have
more to
share
in the
coming
papers
as always
but how
do they
address it
specifically
here
in figure
five
they
map on
we'll root
the tree
so
technically

as shown
right here
it's an
unrooted
tree
there is
an implicit
rooting
here
but it's
a very
interesting
feature
of phylogenetics
that few
outside of the
field know
about
that
computationally
the tree
is often
inferred
in an
unrooted
fashion
like by
clustering
the ones
that are
close
and finding
the
relationships
etc
however
that
is

leaving
an
ambiguity
as to
where
the
root
of
the
tree
is
so
like
for
example
if
this
is
the
overall
connectivity
of
the
based
upon
the
phylogenetic
inference
this is
what
the
tree
topology
has
been
inferred
to
look

like
and
somebody
might
say
well
it
looks
like
this
one
the
red
one
and
the
green
one
below
it
they
are
clearly
very
closely
related
but
actually
if
the root
is
in
here

that's
not
necessarily
the

case
so
the
rooting
of
a
phylogenetic
tree
is
very
important
and
a
tree
that's
rooted
inappropriately
or
has
an
inappropriate
outgroup
selection
is
just
it's
worse
than
illegible
because
it's
highly
legible
and
it
can
have
a

high
statistical
confidence
but
it
can
have
an
absolutely
non-biological
topology
but
assuming
that
they
meant
the
tree
to
be
rooted
here
here
the
branches
are
where
there's
going to
be
different
operators
occurring
including
the
eye
identity
unchanging

one
and
then
the
edges
reflect
like
bifurcations
expetiation
events
so
here
there's
an
ancestral
homeostat
and
then
the
lineage
leading
to
orange
did
not
undergo
any
changes
that
were
non-conservative
on
the
branch
leading
to
these
sister

species
there
was
a
duplication
and
then
it
stayed
the
same
and
stayed
the
same
and
then
the
purple
node
is
going
to
have
two
duplicated
homeostats
and
so
they're
just
overlying
evolutionary
algebra
the steps
that they
gave
onto

the
topology
of
speciation
and
so
that
as
they
discuss
opens
up a
way
to
think
from
this
sort
of
graphical
functional
perspective
which
is
compatible
with
act
inf
to
think
about
how
for
example
from
the
species
today

we
could
infer
some
of
the
schema
of
the
early
vertebrate
brain
or
the
early
primate
brain
or
push
it
back
further
what
about
the
common
ancestors
of
the
vertebrates
and
the
invertebrates
and
all
these
other
questions

but
that's
kind
of
what
they
lay
out
here
and
so
these
are
familiar
models
to
evolutionary
biologists
like
phylogenetic
ancestral
state
reconstructions
of
phenotype
which
are often
even
done
in a
Bayesian
way
like
there's
an
algorithm
beast
Bayesian

estimation
of
ancestral
states
this
is
also
a
Bayesian
model
that
allows
us
to
do
reconstruction
of
ancestral
states
but
it's
in a
very
different
way
than
it's
been
approached
outside
of the
active
world
but that's
figure five
and
that's
where we

see the
evolutionary
algebra
the design
patterns
or the
pattern
language
for
structural
changes
in
generative
model
superimposed
on
species
relatedness
in a
phylogenetic
tree
any
thoughts
on five
i tried
to convert
this and
reapply it
to my
picking of
teams in
the march
madness
table and
i wasn't
able to
take
phylogenetic

trees and
convert them
into winning
thousands of
dollars on
betting on
college basketball
so despite
my best
efforts
it's
it's kind
of contained
and for
what it
what it
represents
i mean not
to foreshadow
too hard but
i think
live stream
number 39
and 40
might constitute
march
madness
and then
also here's
another um
you know
for those
who watch
okay here
now we
have a little
bit of a
march

madness
bracket
going don't
we here's
bracketology
that is
so fantastic
so here
we go
yeah so
now we
have um
now here
the root
could have
been here
and then
here's like
two big
species
clades
like two
like this
is like
vertebrates
and invertebrates
and then
there's some
rooted clade
wow
so that's
like why it's
important to
root the tree
because here
it looks like
all of these
are sister to

each other
at the
exclusion of
this one
but then if
we were to
have rooted
that tree
like here
then actually
this is a
small sister
outgroup to
all of
these
right
and so
the rooting
and the
contextualizing
is really
important
it's just
like it's
super interesting
because it
seems like
you put in
so much
data into
these
phylogenetic
models and
it seems
like literally
how could
this not
converge on

a super
obvious
answer
i mean
isn't it
clear how
these answer
related to
each other
phylogenetically
but then
there's a
little bit
of nuance
that enters
the picture
steven
yeah i was
curious that
the allostatic
can
recombine
with the
homeostatic
to give a
base level
temporal
so it's
um
like that
doesn't
it doesn't
deflate
back to
just staying
allostatic
i suppose
there would

be they
just don't
show that
it could
um
it could
have
if it
did that
i suppose
you just
wouldn't
talk about
it
you know
it's got
a
when it
goes a
then it
goes a
i
goes to
ht
so yeah
it's a
th
yeah so
we have
a homeostat
here we
have two
homeostats
two homeostats
and now
here we
have one
homeostat

and one
allostat
in this
blue
node
and so
that
um
but
they're just
kind of
giving general
examples
like it
would have
been cool
to see
um
this is
where we're
talking about
um
mammals
and here's
non-mammal
vertebrates
like here
is a
specific
brain region
and also
that is
in this
um
chakra
borti
and jarvis
2015

paper
this
paper
um
gets
more into
the nuances
of neuroanatomy
and neurogenetics
and how
like duplication
in
um
brain regions
can be
underpinned
by developmental
neurogenetic
changes
and they do
link that a
little bit more
closely like to
studying the
bird brain
and the
song and
motor
system
in parrots
which is also
cool because
birdsong has been
studied with
act-inf so
like there
could be a
nice

building upon
this
work
and connecting
it to some
of the
um
structure
fitting
in figure
five
so it
could almost
be like
you got
yeah you
got parts
of the
brain
which
are more
allostatic
orientated
or
homeostatic
orientated
and
that so
in a way
by
the
structuring
of that
gives
some sort
of
differentiation
in terms

of how
information
and
control
would be
carried
out
yeah
totally
and like
um
extending
temporal
depth
within
a model
it's like
a brain
region
getting a
little
bigger
let's
just
say
not that
size
is correlated
with function
in that
specific
way
but then
duplicating
brain
regions
is like
duplicating

laterally
or nesting
a brain
region
it's
hierarchical
think about
how the
eyes
developed
the insect
eye
it has
a relatively
simple
module
but then
insect
eyes
can change
in size
over
evolutionary
time
very radically
from like
taking up
you know
half of the
head
to being
just one
photoreceptor
or even
being lost
because
they're
existing

in more
of a
duplicable
motif
whereas
the binocular
eye system
that mammals
have
it's
not as
amenable
to like
just
doubling
into
four
eyes
or
you know
splitting
into two
in the
middles
or something
like that
whereas
the insect
compound
eye
has like
some of
those
evolutionary
affordances
so then
understanding
parameter

change
within a
model
and then
structure
learning
on
populations
of models
and what
are the
mutational
adjacencies
there's
so many
awesome
areas
and it's
like truly
just beginning
for evolutionary
bio and
active
stephen
yeah it was
really helpful
like having
someone like
yourself who's
got that
background
to go through
that's quite
helpful because
it's a
it's a little
it's a little
bit intimidating

actually
at first sight
seeing this
for me
but I and I
see the logic
what you're saying
it's quite
exciting
actually seeing
that
again one of
the things
this paper
does is it's
it's tying
threads together
so which
seem very
distant
almost in
traditional
terms like
well I don't
like using the
word kamikaze
but it's a
little bit
like yeah
it's a bit
overwhelming
often so I
think it's
but of course
that's the
beauty of
Act-Inf
is it has

that unifying
potential
so thanks
cool yeah
thank you
I agree
like there's
some labeling
this is rarely
included unless
it's a time
calibrated
phylogenetic
tree but
like we're
looking at
the passage
of time
but that's
not even
always shown
and so
yes Dean
we'll have
our final
thoughts
can you just
go back to
that because
I really
I didn't prompt
you to put
that up there
but I really
like that
because
this at
some point

we're gonna
have to
in point
to talk
about
that sort
of basal
gating part
of this
and most
things are
are
are
organized
as if
then causality
but you just
said it
it's then
if
go or don't
go
is very
dependent
on then
before
we choose
to go
or don't
go
right
go or
don't
go is
still
held
open
until

one
of the
others
decided
so
another
thank you
for that
because
that
again
that
in the
point
two
we
should
maybe
talk
a little
bit
about
that
cool
yeah
there's
so many
ways
to think
about
like
this
garden
of
forking
paths
branching
paths

how
that
relates
to
parameter
updating
Bayesian
belief
updating
and
structure
learning
what
about
when
the
structure
of
a
model
is
a
parameter
in
a
nested
model
so
then
that
sort
of
blurs
the
line
from
the
bottom

looking

up

it

looks

like

structure

learning

but

then

from

the

top

looking

down

it

looks

like

parameter

fitting

right

and

if

only

we

had

like

some

mathematics

!

to

describe

that

okay

any

final

thoughts

or

what

are
we
excited
about
for
the
dot
two
next
week
well
my
last
thought
is
yeah
I'm
excited
to keep
this
going
because
I
think
it's
got
a lot
of
legs
to
it
I
think
one
thing
that
comes
to

mind
as
well
with
all
this
progressively
increased
temporal
and
hierarchical
depth
is
maybe
where
some
of
the
more
basic
elements
still
have
a
really
useful
part
to
play
is
what
Mike
Levin
said
when
do
we
stop

we
got
all
these
ways
we
can
think
how
do
we
know
when
to
stop
and
sit
down
maybe
that's
where
the
gut
is
so
useful
at
the
end
of
the
day
the
brain
can
go
running
off

and
maybe
the
gut
and
the
heart
need
to
say
okay
sit
down
get
some
food
so
I
might
well
I'm
going
to
get
a
drink
of
time
but
I'll
hear
your
last
thoughts
as
well
Dean
I

hate
having
the
last
word
so
you
won't
have
it
don't
worry
you'll
have
it
and
I
just
appreciate
what
Steven
just
said
because
I
think
that
that's
a
big
part
in
this
ability
to
go
from
word

back
into
the
phenomenological
space
and
I
think
there
are
other
brains
that
we
know
of
now
that
play
a
significant
role
in
that
not
just
the
one
that
turns
everything
into
symbols
great
my
last
thought
is

there's

no

better

drink

after

a

foraging

trip

than

honey

all

right

thank you

and talk

to you

later

bye

bye